Prática com LLMs

Prof. Dr. Hendrik Macedo
Departamento de Computação

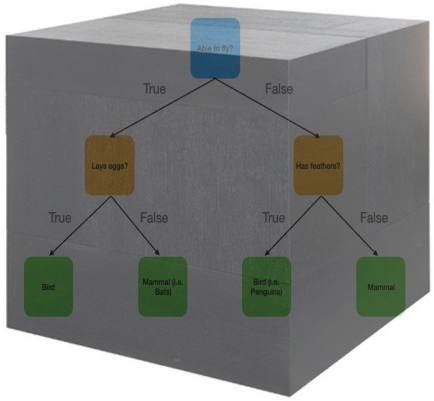
Modelos Discriminativos

- Focam em **limites de decisão** para separar classes.
- Modelam a **probabilidade condicional**:

$$P(Classe|Dados)$$

ID	Animal (Nome)	eggs?	fly?	feathers?	Class
1	Sparrow (Pardal)	True	True	True	Bird
2	Eagle (Águia)	True	True	True	Bird
3	Parrot (Papagaio)	True	True	True	Bird
4	Penguin (Pinguim)	True	False	True	Bird
5	Ostrich (Avestruz)	True	False	True	Bird
6	Bat (Morcego)	False	True	False	Mammal
7	Dog (Cachorro)	False	False	False	Mammal
8	Cat (Gato)	False	False	False	Mammal
9	Pigeon (Pombo)	True	True	True	Bird
10	Elephant (Elefante)	False	False	False	Mammal
11	Hawk (Falcão)	True	True	True	Bird
12	Lion (Leão)	False	False	False	Mammal
13	Kiwi (Kiwi)	True	False	True	Bird
14	Whale (Baleia)	False	False	False	Mammal
15	Platypus (Ornitorrinco)	True	False	False	Mammal

modelo



$$P(Bird?\,|True, False, True)$$

Modelos Generativos

- Focam na **distribuição dos dados**
- Modelam a **probabilidade conjunta**:

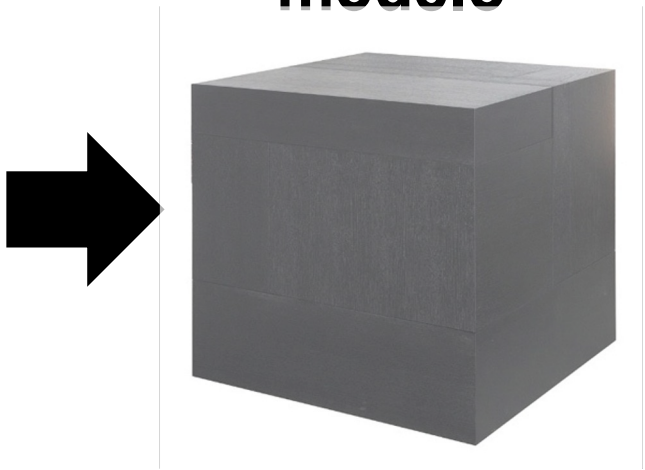
$$P(Dados, Classe) \text{ ou } P(Dados).$$

- Assume um processo de geração:

$$P(Classe) \rightarrow P(Dados|Classe)$$

ID	Animal (Nome)	eggs?	fly?	feathers?	Class
1	Sparrow (Pardal)	True	True	True	Bird
2	Eagle (Águia)	True	True	True	Bird
3	Parrot (Papagaio)	True	True	True	Bird
4	Penguin (Pinguim)	True	False	True	Bird
5	Ostrich (Avestruz)	True	False	True	Bird
6	Bat (Morcego)	False	True	False	Mammal
7	Dog (Cachorro)	False	False	False	Mammal
8	Cat (Gato)	False	False	False	Mammal
9	Pigeon (Pombo)	True	True	True	Bird
10	Elephant (Elefante)	False	False	False	Mammal
11	Hawk (Falcão)	True	True	True	Bird
12	Lion (Leão)	False	False	False	Mammal
13	Kiwi (Kiwi)	True	False	True	Bird
14	Whale (Baleia)	False	False	False	Mammal
15	Platypus (Ornitorrinco)	True	False	False	Mammal

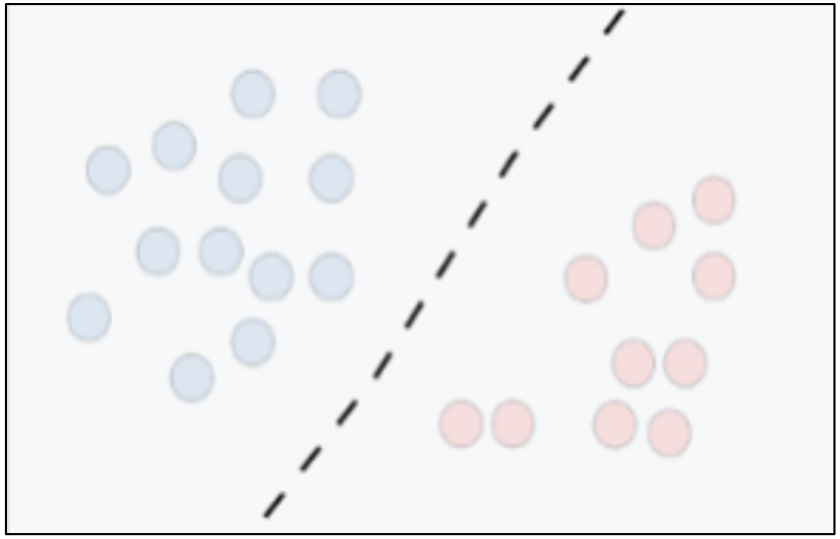
modelo



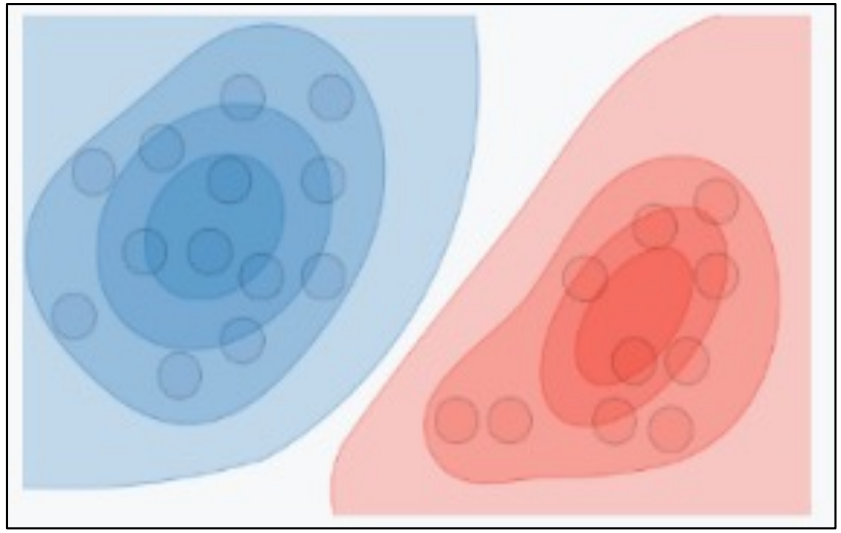
$$P(True, False, True|Bird)$$

Modelos discriminativos vs generativos

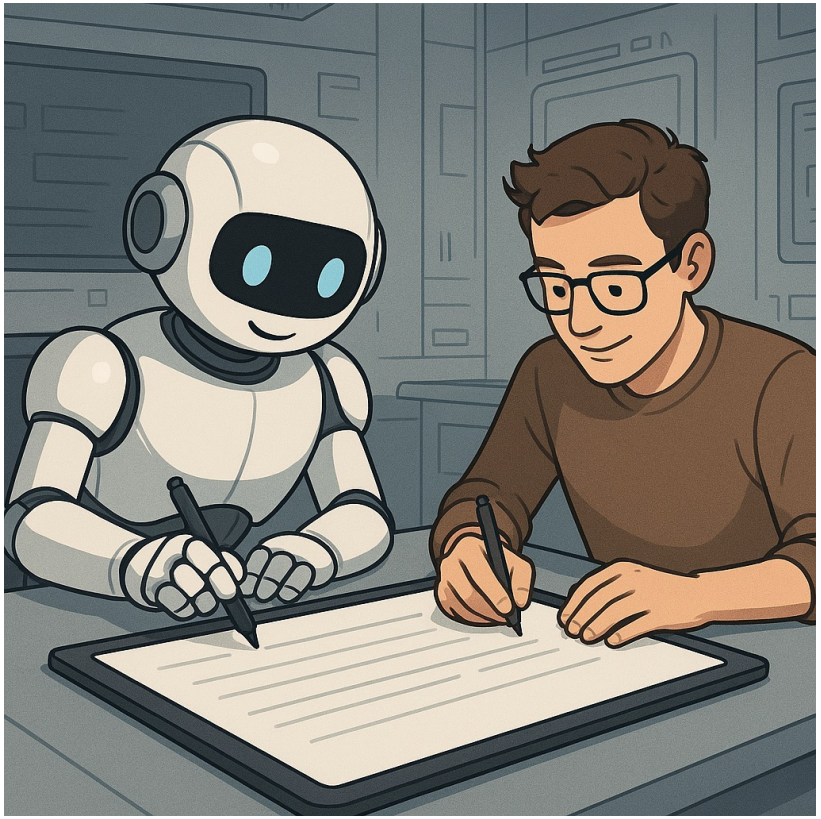
Modelos discriminativos



Modelos generativos



Sobre como a máquina representa a linguagem humana



Fundamentos do N-Gram

- A forma mais simples de Modelo de Linguagem estatístico.
- ...

Tokenização

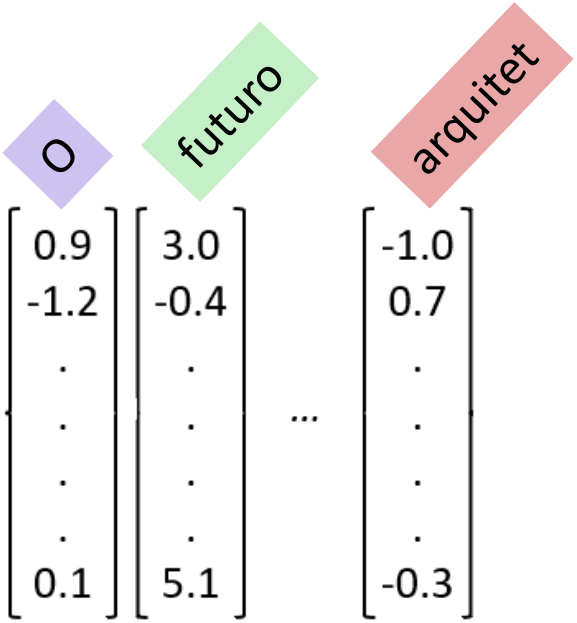
```
O futuro da IA em PLN (Processamento de Linguagem Natural) baseia-se em arquiteturas neurais que superam as limitações de contexto fixo e escassez de dados herdadas dos N-Grams.
```

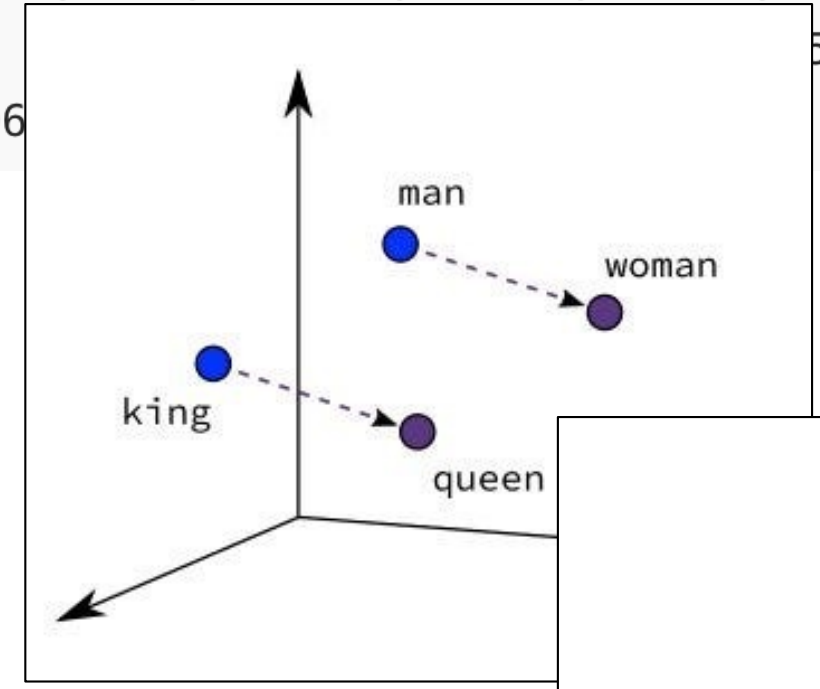
```
[46, 28600, 1033, 59199, 863, 177082, 350, 7212, 4626, 334, 126554, 8551, 23735, 8, 3611, 535, 5460, 863, 123148, 12558, 36501, 1873, 661, 2539, 313, 472, 117731, 4693, 334, 50124, 9295, 78, 319, 4575, 5788, 89, 334, 24315, 94705, 3921, 3406, 478, 85061, 2247, 2517]
```

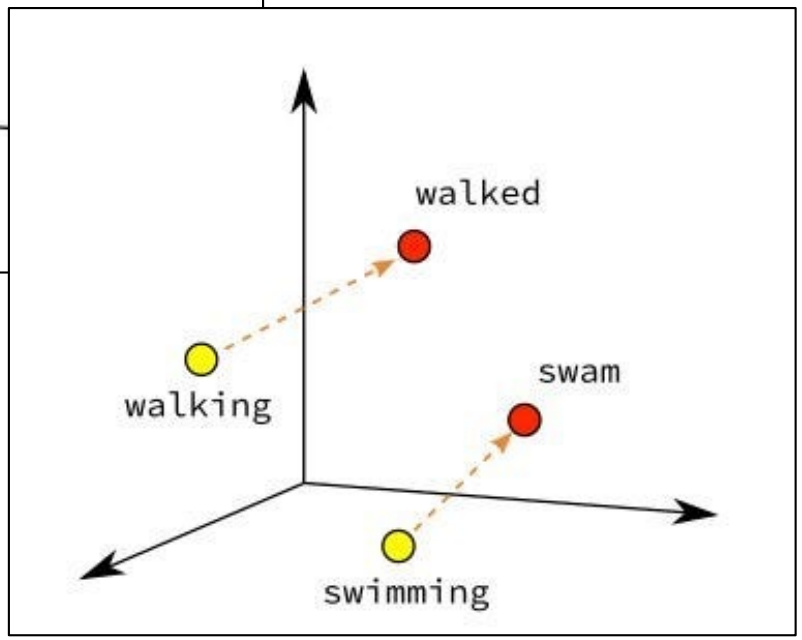
Embeddings (representação vetorial)

```
[46, 28600, 1033, 59199, 863, 177082, 350, 7212, 4626, 334, 126554, 8551,
23735, 8, 3611, 535, 5460, 863, 123148, 12558, 36501, 1873, 661, 2539,
313, 472, 117731, 4693,                                  5, 5788, 89, 334,
24315, 94705, 3921, 3406
```

embeddings

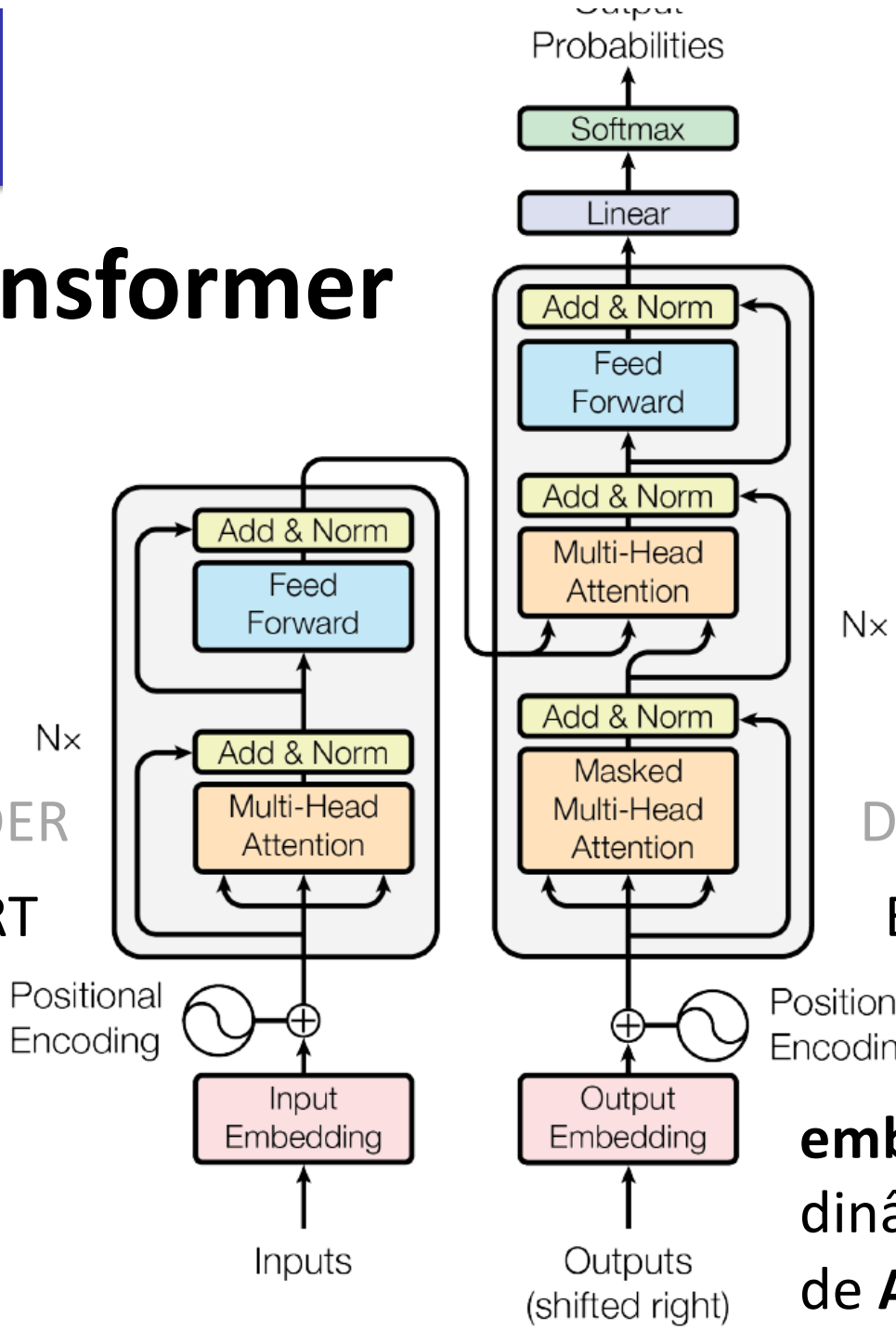






Transformer

Vaswani, Ashish, et al. **"Attention is all you need."** *Advances in neural information processing systems* 30 (2017).



ENCODER

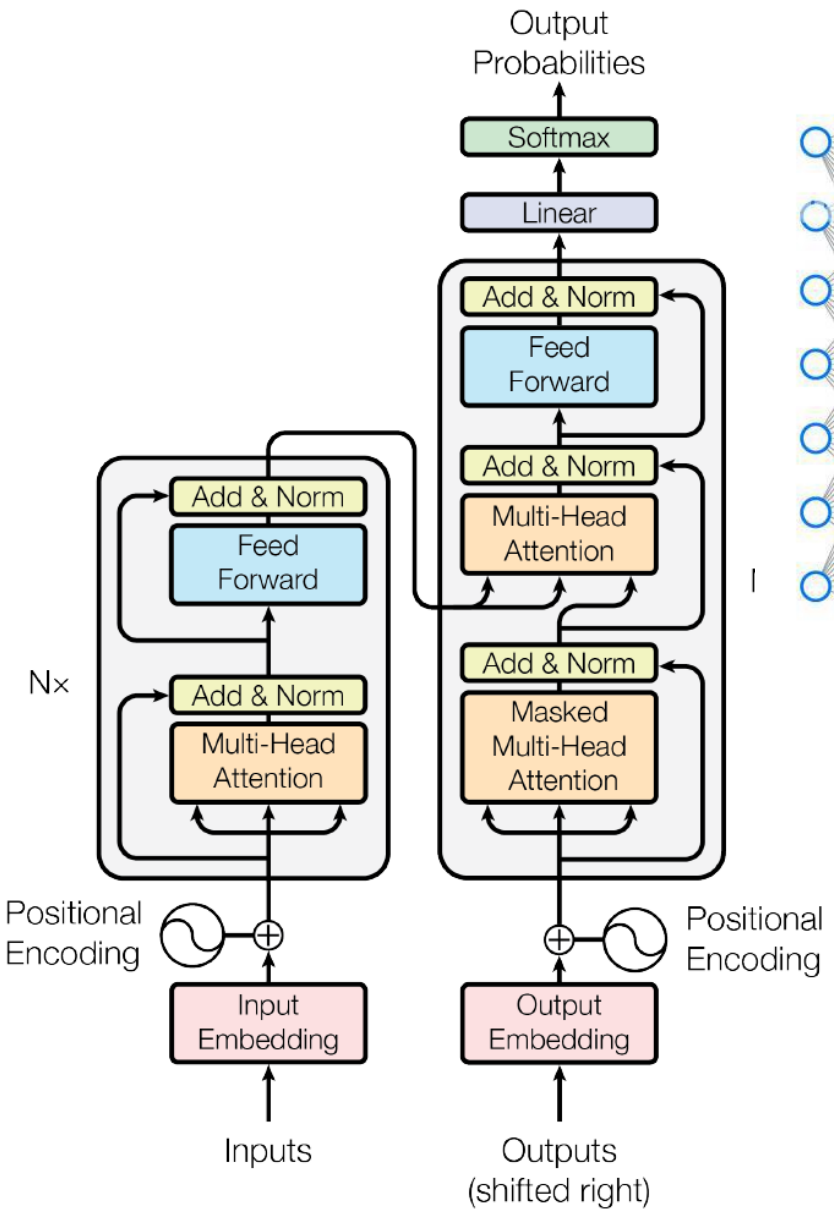
Ex: BERT

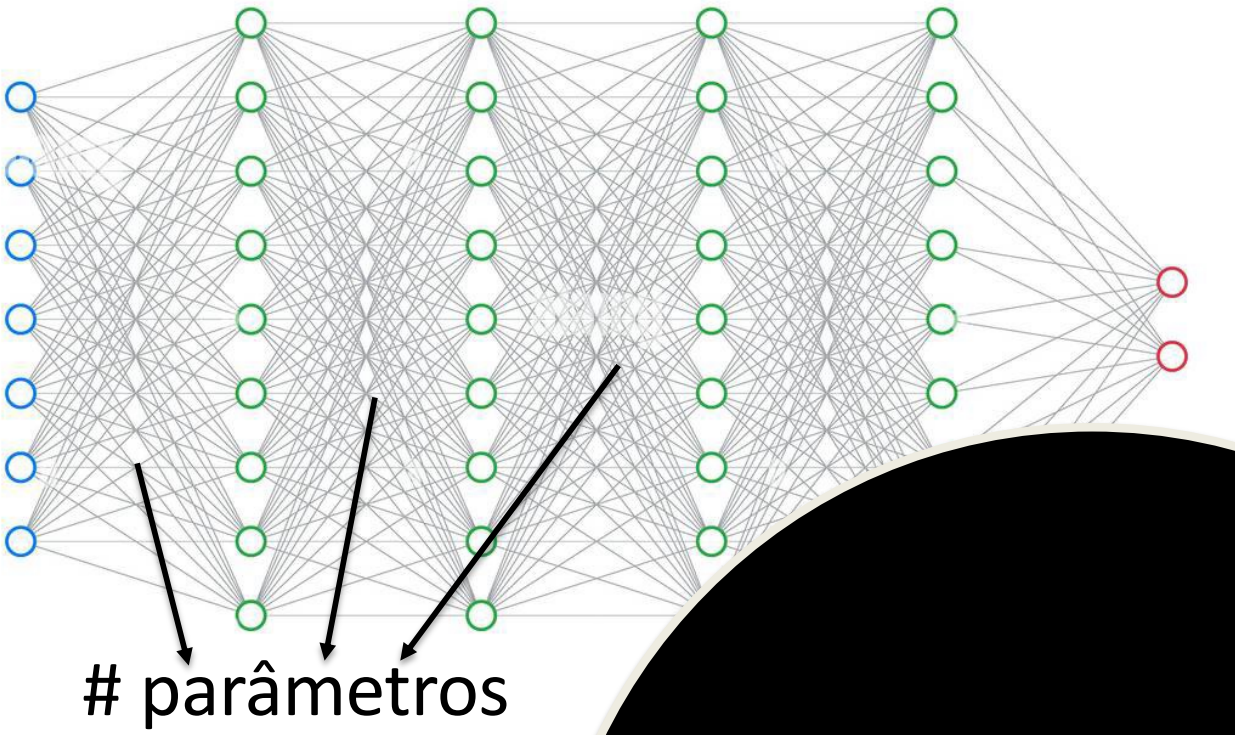
DECODER

Ex: GPT

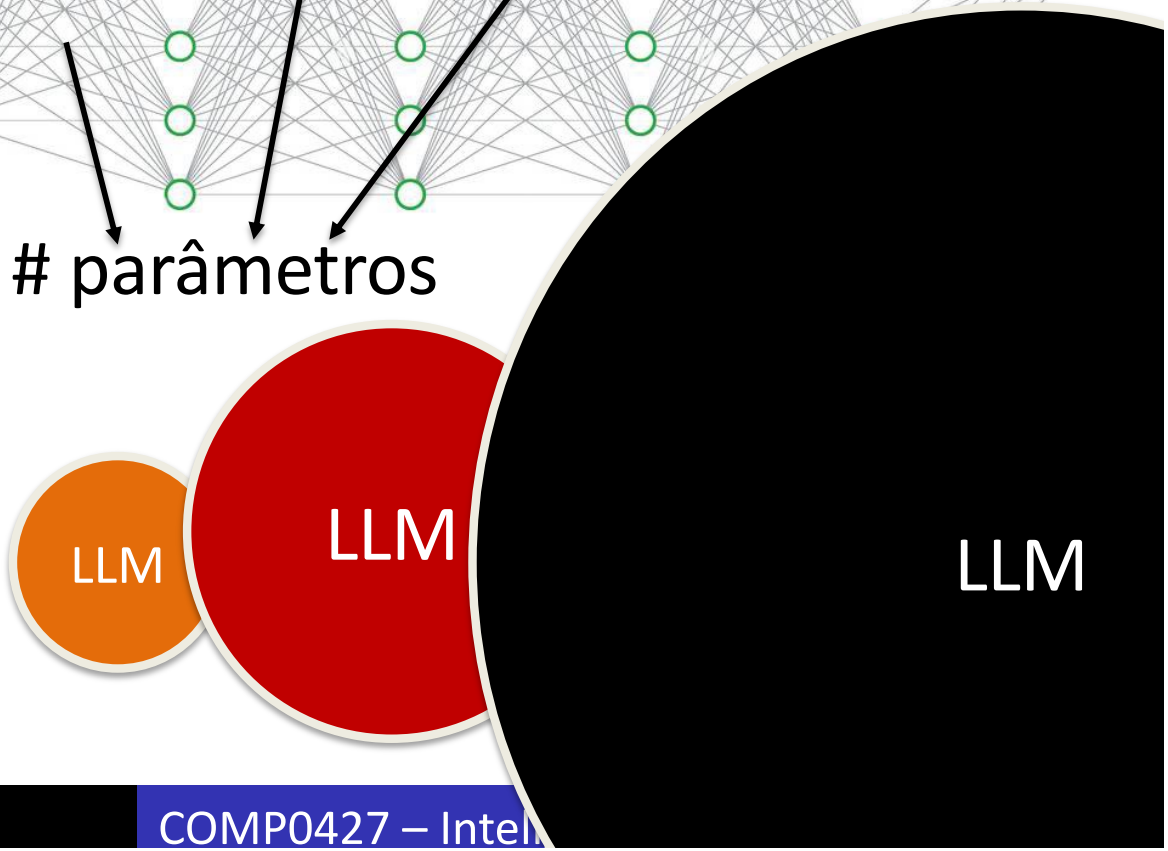
embeddings contextuais (ou dinâmicos) através do mecanismo de **Auto-Atenção** (*Self-Attention*)

Transformer

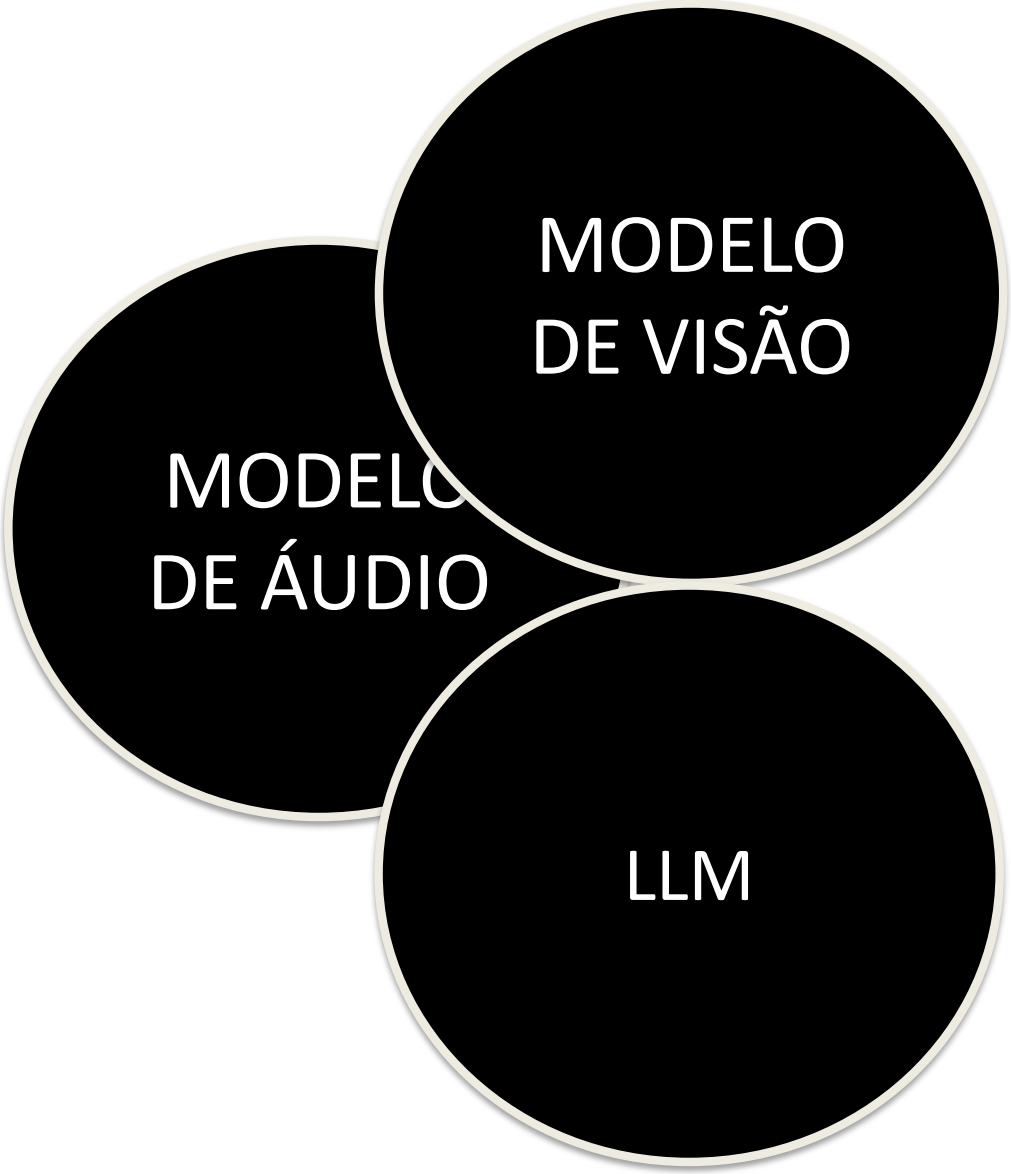


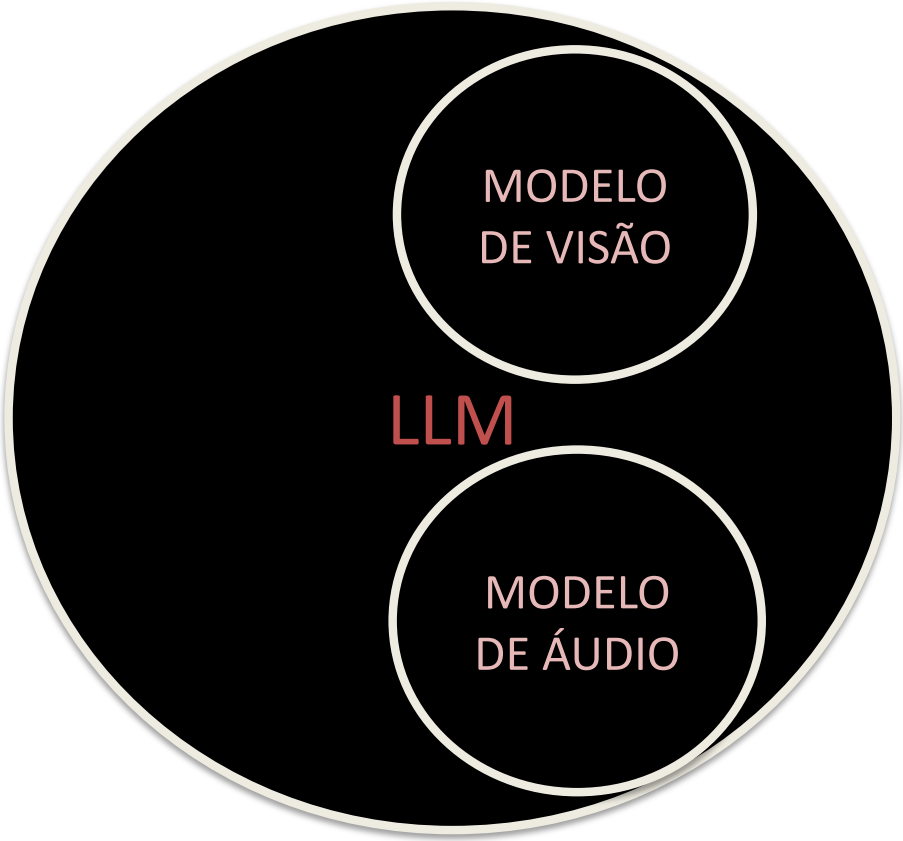


parâmetros

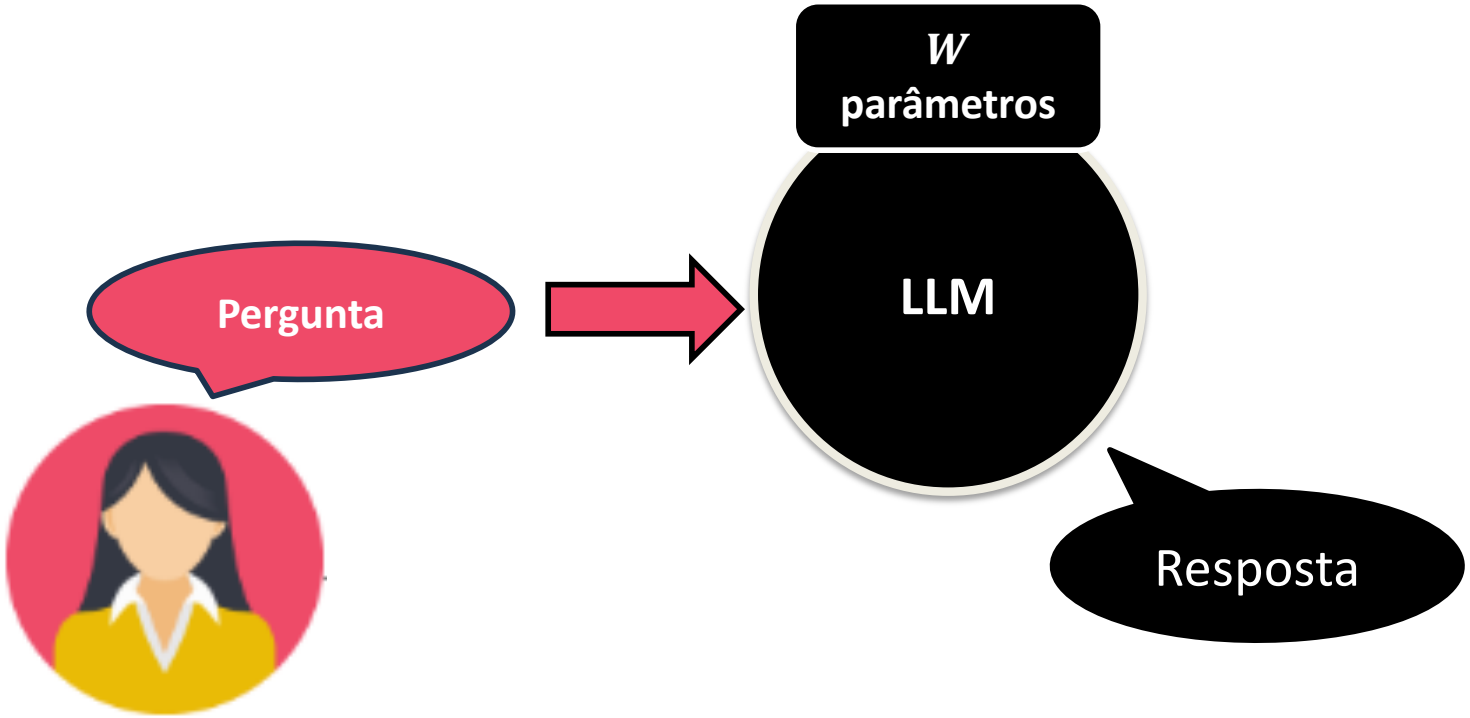


Multimodalidade

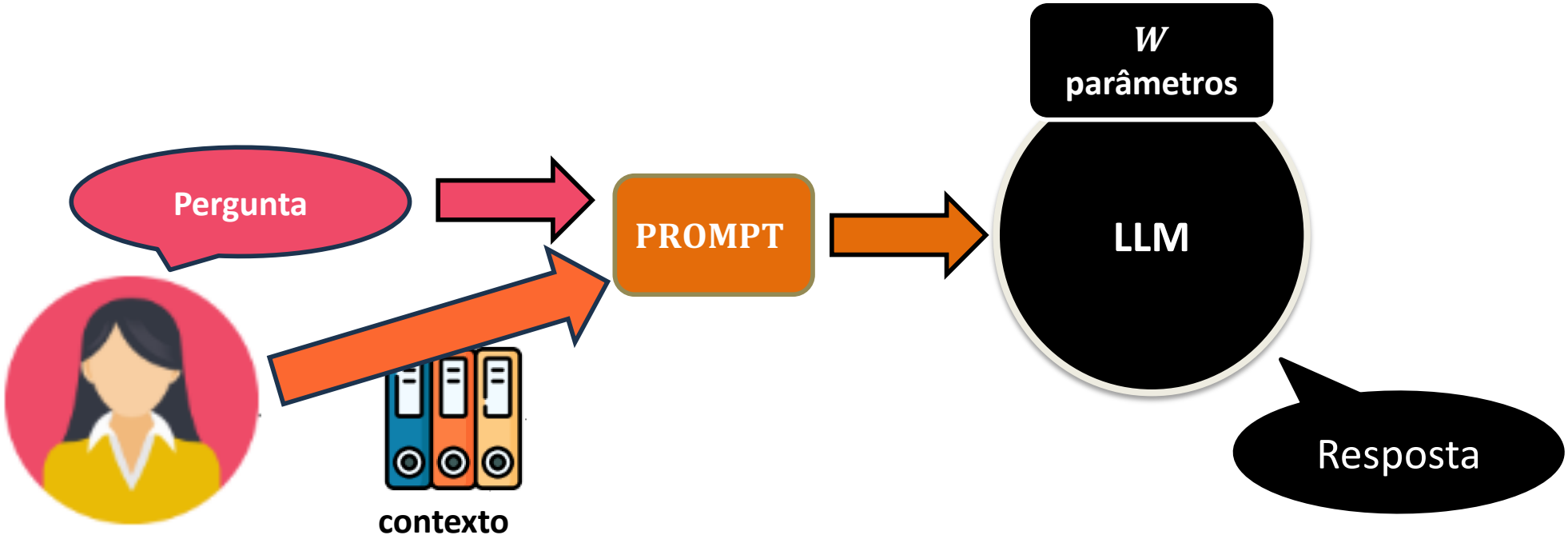




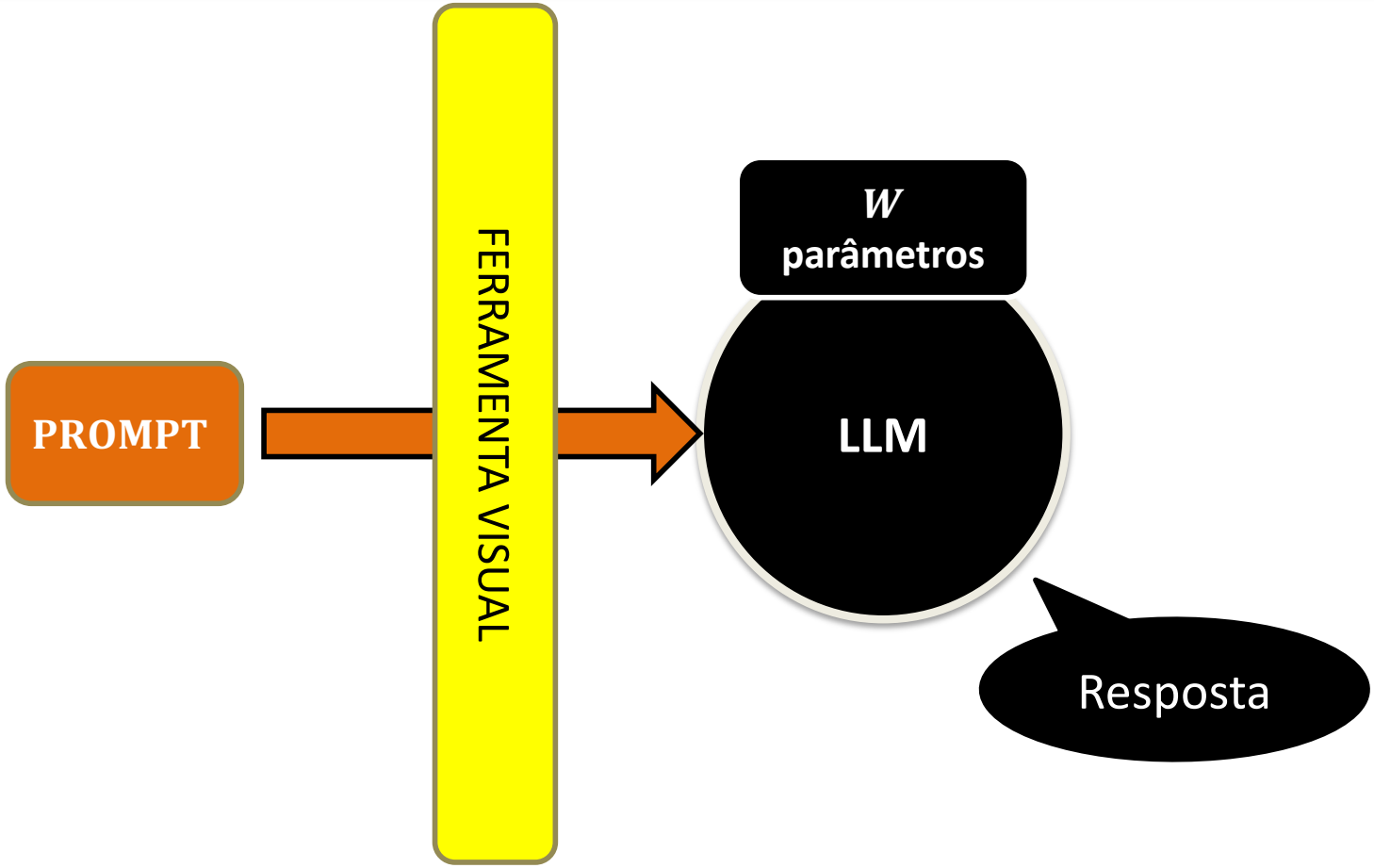
Uso



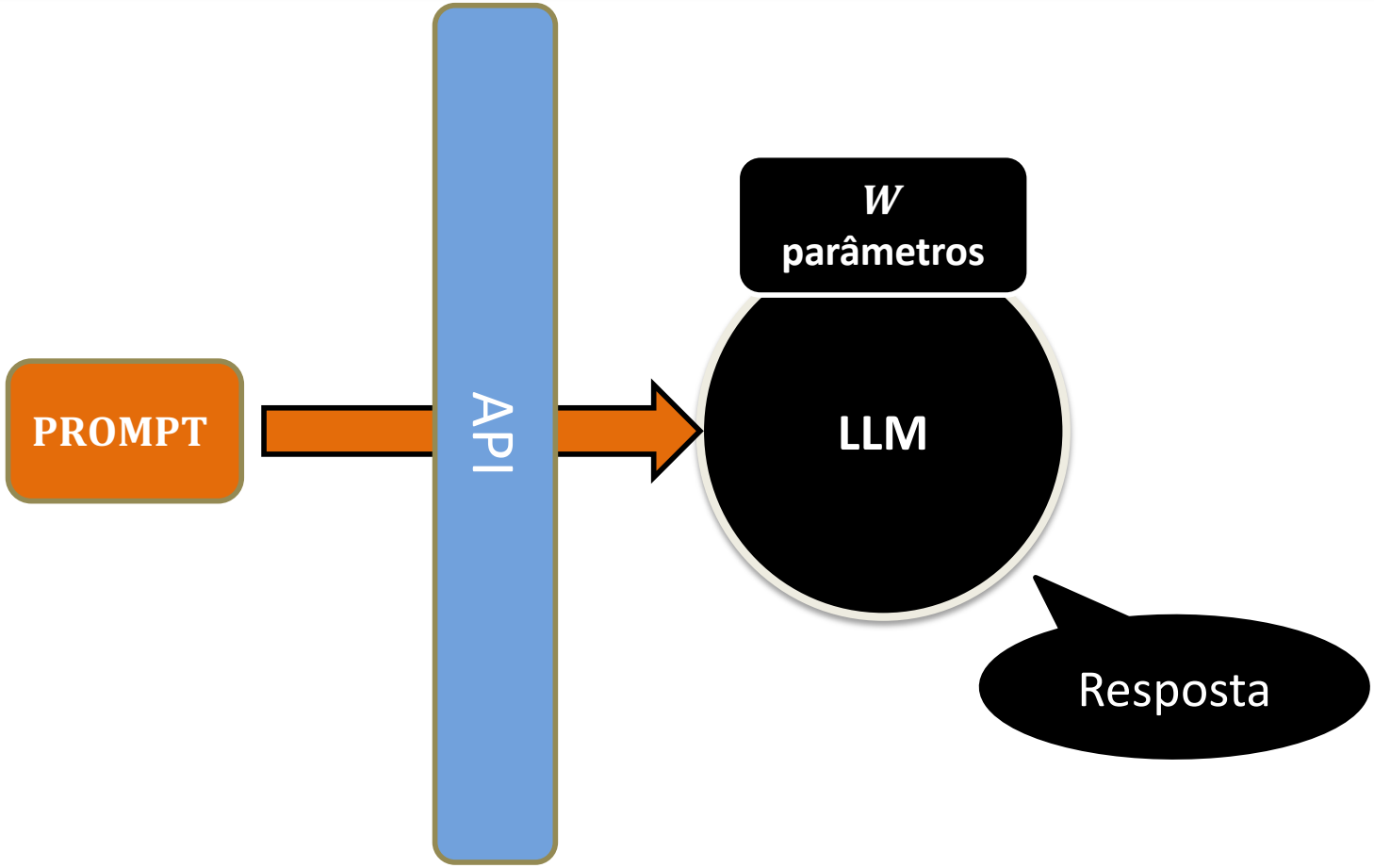
Uso



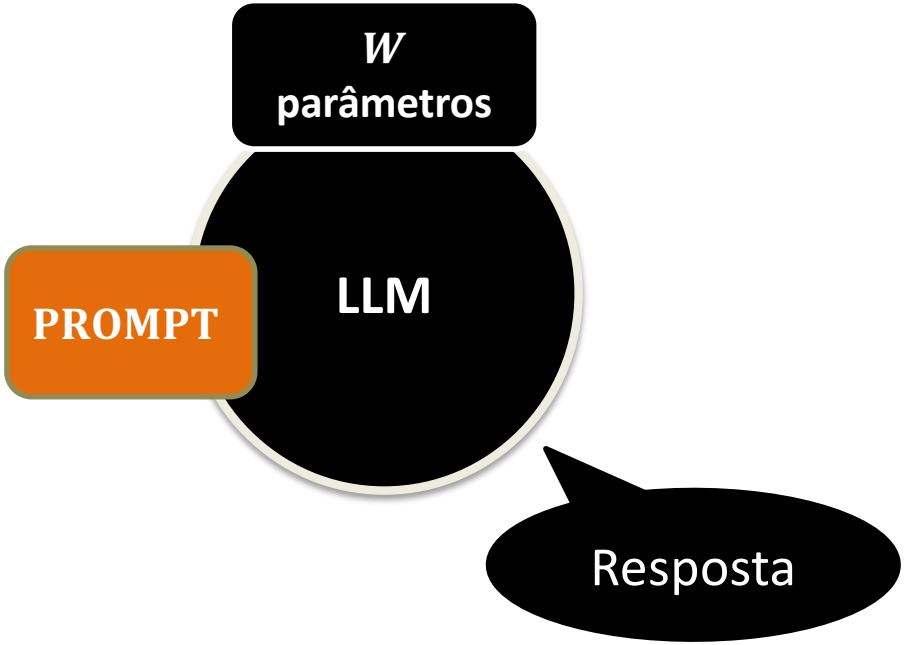
Uso via software



Uso programativo via API



Uso programativo local



Notebooks Colab

- Texto: https://colab.research.google.com/drive/1XiGPhifSozjxlREs9OoYgexTs5D557no?usp=sharing
- Imagem: https://colab.research.google.com/drive/1BsWVE14D1h2W1lnXN4Rpgrv4aVlzAPVl?usp=sharing
- Áudio: https://colab.research.google.com/drive/1nDbbhFEXeCcLLKpO3Of-HJlmi1F2Uyhi?usp=sharing